

Reconstructing sparse river dissolved organic carbon with environmental prediction and observation-aware residual learning

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Abstract

Sparse and irregular monitoring limits the reconstruction of river dissolved organic carbon (DOC), particularly when observations are missing for extended periods or entire stations. Environmental machine learning supplies strong background predictions, but its errors can retain information associated with the available observation history. We propose a hybrid reconstruction framework that combines an environmental-context tree predictor with an observation-aware temporal residual expert. The latter learns station-blocked out-of-fold errors of a local tree model using ecological features, observation age, support indicators, and recurrent memory. A validation-fitted combination integrates the two experts for temporal gaps, while a context-only configuration serves random gaps and unmonitored stations. Across a 357-station river cohort and five training seeds, the temporal hybrid reduces MAE from 1.047 to 0.887 mg/L for unobserved periods and from 1.016 to 0.784 mg/L for observation-assisted periods, improvements of 15.3% and 22.9% over the selected ExtraTrees context predictor. It also improves over the complete temporal residual expert, supporting complementary prediction roles. A separate spatial-adaptation experiment shows that five observations per target station reduce regional-model MAE by 17.9% on a fixed query. These results establish the value of combining environmental prediction with residual temporal learning for sparse DOC reconstruction, while identifying cross-station transfer and high-concentration events as priorities for further development.

Keywords: dissolved organic carbon; sparse monitoring; water-quality reconstruction; residual learning; recurrent neural network; spatial adaptation

1 Introduction

River dissolved organic carbon connects terrestrial carbon sources with downstream aquatic environments. Its concentration varies with watershed properties, hydrological conditions, and biogeochemical processing, with substantial differences among regions [1]. Reconstructing DOC records can support analysis of long-term carbon dynamics and comparison among catchments. Recent work has demonstrated the value of combining machine learning with environmental and satellite information to recover river organic-carbon patterns over large areas and extended periods [2]. These applications require predictions at locations and times where direct chemical measurements are unavailable.

The observation problem is persistent. Water chemistry is often sampled less continuously and less consistently across sites than meteorology or streamflow, as illustrated by large-sample chemistry compilations [3]. A river monitoring record can contain isolated gaps, prolonged interruptions, partially observed later periods, and stations with little or no target chemistry. Each case leaves a different set of information available for reconstruction. A nearby measurement may help with

a short gap, whereas an unmonitored station must rely more heavily on environmental similarity and observations elsewhere.

Existing methods provide complementary strengths. Temporal interpolation and seasonal summaries offer simple local references when surrounding observations exist. Stream-network geostatistics represent hydrological distance and flow connectivity, extending spatial prediction beyond straight-line proximity [4]. Environmental machine learning, including random forests and extremely randomized trees, can represent nonlinear associations among predictors [5, 6]. Such models can also use manually constructed lags and neighboring-observation summaries; their predictive ability is therefore a demanding baseline for a new temporal method. Recurrent graph models already combine spatial and temporal representations for missing-data reconstruction [7], while observation-aware recurrent models use missingness masks and elapsed observation time to inform memory updates [8].

The remaining methodological question concerns how these information sources should share the reconstruction task. A strong environmental predictor can explain much of the background variation, leaving errors that may depend on observation history. Asking a neural model to predict the complete DOC signal requires it to relearn that background as well as any remaining dynamics. Conversely, a tree model supplied with a fixed set of lags may not capture all useful relationships in an irregular observation sequence. This motivates a division of prediction into an environmental reference and a temporal correction, followed by a combination that can retain useful information from both.

We develop an environmental-context and observation-aware residual reconstruction framework for DOC. A fixed tree model provides a local reference for training a recurrent residual expert. Station-blocked out-of-fold predictions define its training targets, so residual learning is based on predictions at withheld training stations rather than the tree model’s in-sample fit. Observation age, recent support, ecological descriptors, and hydroclimatic inputs inform the recurrent correction. A separate environmental-context expert supplies a strong prediction using visible local and river-network summaries, and the two complete expert predictions are combined on validation data.

This study makes three contributions. First, we introduce a hybrid DOC reconstruction method that combines environmental prediction with observation-aware temporal residual learning, trained on station-blocked out-of-fold errors. Second, we demonstrate the complementary value of its environmental and temporal predictors through comparisons with strong tree baselines and matched component experiments across distinct monitoring gaps. Third, we develop and evaluate a sparse station-calibration extension, showing how a small local sample can improve regional predictions at poorly monitored stations. Together, these contributions connect a practical reconstruction method with evidence about the information needed to recover sparse DOC records.

2 Materials and methods

2.1 River cohort and reconstruction tasks

We use a Mississippi River basin cohort of 357 stream stations connected by 324 directed river-network edges. The monthly grid spans April 1972 to September 2026, comprising 654 months and 233,478 station-month positions. Only 22,571 positions (9.67%) contain observed DOC, illustrating the sparsity of the reconstruction problem. The present study focuses on DOC concentration in mg/L. Dynamic covariates are water temperature and discharge with availability indicators. Static inputs describe station location and ecological regime. DOC targets are transformed as $z = \log(1 + y)$, where y denotes concentration.

The four evaluation scenarios represent different monitoring gaps (Table 1). Random gaps withhold observed station-month cells. Unobserved periods withhold target observations in a later time segment. Observation-assisted periods retain designated observations as context in that segment. Unmonitored stations withhold the target record at a group of stations. We use one existing split per scenario and five training seeds, 42–46. Splits are held constant across the compared methods. Spatially and temporally structured evaluation follows the principle of matching a split to its intended prediction setting [9].

Table 1: Evaluation tasks. Counts are unique query cells and contributing stations within each scenario. The same cells can occur in different scenarios.

Scenario	Query cells	Stations
Random gaps	4,514	355
Unobserved periods	2,223	69
Observation-assisted periods	1,778	67
Unmonitored stations	2,531	43

During model fitting and validation inference, target inputs contain only training and designated context observations. Validation labels enter early stopping and model combination, but not the input features. Terminal inference additionally permits validation observations as context; the evaluated query labels remain hidden. Current hydrological covariates are assumed available. The task is reconstruction of missing DOC under specified information availability, rather than forecasting future hydrology.

2.2 An environmental-context predictor

The context predictor C_{it} uses 39 features for station i and month t : temperature and discharge with their visibility flags, seasonal sine/cosine terms, coordinates, 13 regime descriptors, six visible-DOC context summaries, and 12 local lag features. The context summaries contain global, upstream, and downstream visible DOC means and support fractions. Means are calculated in log-transformed target space and exclude the query cell. The local lags describe observations at one, three, six, and twelve months, including availability.

All context forests contain 300 trees. For each seed and scenario, validation MAE selects among six candidates: random forests with minimum leaf sizes one or two, and ExtraTrees variants using all-feature or square-root-feature split selection and minimum leaf sizes one, two, or four. All selected context experts are ExtraTrees: leaf size four in the temporal scenarios, leaf size two for random gaps, and leaf sizes one or two for unmonitored stations. A fixed 300-tree random forest with leaf size one is retained as an additional baseline on the same features, seeds, and query cells.

This predictor already uses explicit temporal and river-context information. Its comparison with the hybrid therefore measures the added value of the learned residual expert and combination, beyond an environmental model that already has engineered history and neighborhood features.

2.3 An observation-aware temporal residual expert

2.3.1 Local reference and out-of-fold residual targets

A separate local forest defines the reference for residual learning. It is an ExtraTrees model with 300 trees, minimum leaf size four, and all features available at each split. Its 40 predictors retain

environmental and lag information but replace the six network-context columns with seven strictly historical local features: the most recent value, log observation age, an availability flag, and observation counts over the preceding one, three, six, and twelve months. This local reference B is distinct from the selected context expert C .

We partition training stations into five folds. For each fold, the local forest is fitted on the other four folds, and all target observations at the withheld stations, including context observations, are hidden from its input construction. Let B_{it}^{OOF} be the resulting prediction in log space. The neural target is

$$r_{it}^{\text{OOF}} = \frac{z_{it} - B_{it}^{\text{OOF}}}{\sigma_{\text{train}}}, \quad (1)$$

where σ_{train} is the standard deviation of transformed training labels. The neural branch uses the corresponding fold-specific observation view. This construction gives the branch an error signal from station withholding rather than residuals from an in-sample forest fit.

2.3.2 Observation-aware memory

The neural branch combines a two-layer ecological encoder with a two-layer, 64-unit node encoder. The ecological encoder maps nine regime variables through two 32-unit layers. Monthly node inputs contain hydroclimatic values and masks, season, location, visible DOC and its availability, hydro-regime descriptors, and observation-support features. The latter include the last visible value, log age, an ever-observed flag, support fractions at four time scales, and upstream support. Downstream support is zero in the selected residual configuration.

For each prediction month, a 12-month window is processed by an observation-aware GRU with 64 hidden units. Inspired by observation-decay approaches [8], the hidden state is attenuated before its recurrent update:

$$\gamma_{it} = \exp\{-\text{ReLU}(W_d q_{it} + b_d)\}, \quad (2)$$

$$h_{it} = \text{GRUCell}([e_{it}, v_t], \gamma_{it} \odot h_{i,t-1}), \quad (3)$$

where e_{it} is the encoded monthly state, v_t indicates valid history, and q_{it} contains scaled log observation age, an observation-availability indicator, upstream support, and a zero downstream-support channel. Each window starts from zero recurrent state, and padded positions make no recurrent update. Input windows end at the prediction month. Visible observations at the current month may enter support features; hidden query values cannot.

The residual expert uses a self-node encoder without learned edge messages or an attention operator. River connectivity enters through explicit observation-support features. Separate message and lag-attention controls examine the potential additional value of learned river propagation.

2.3.3 Residual prediction and training

A zero-initialized scalar head maps the final hidden state to a standardized residual $d_{\theta,it}$. Using a local forest refitted on all training cells, the complete residual-expert prediction is

$$R_{it} = \max\{0, \exp(B_{it} + \sigma_{\text{train}} d_{\theta,it}) - 1\}. \quad (4)$$

At initialization, the correction is zero and this expert equals its local reference. The forest is fixed throughout neural training.

Training minimizes mean squared error on the standardized out-of-fold residuals. Five station folds contribute to one accumulated optimizer step per epoch. We use Adam with learning rate

10^{-3} , gradient-norm clipping at one, dropout 0.1, at most 20 epochs, and early-stopping patience five. Completed temporal runs stop after 15–16 epochs. Early stopping evaluates the complete local-reference-plus-residual prediction. Target and dynamic-input scaling are based on training cells; static descriptor scaling uses the known station cohort. The prediction applies a lower bound of zero after inverse transformation, with no upper-quantile clipping.

2.4 Combining the two complete experts

For temporal scenarios, the candidate outputs are context-only, residual-expert-only, a geometric blend, and a log-affine combination. The latter is

$$\hat{z}_{it} = \beta_0 + \beta_C \log(1 + C_{it}) + \beta_R \log(1 + R_{it}), \quad (5)$$

$$\hat{y}_{it} = \max\{0, \exp(\hat{z}_{it}) - 1\}. \quad (6)$$

Unconstrained least squares fits the affine coefficients on validation predictions. The geometric-blend weight is selected on a grid with step 0.005. Raw-scale validation MAE selects the candidate output. All ten temporal seed–scenario combinations select the affine form. Its weights combine complete expert predictions and are neither attention weights nor physical coefficients.

Random gaps and unmonitored stations use the context expert directly under the existing scenario-specific configuration. These two routes are specified by the implementation, rather than inferred by a learned routing network. Their final outputs therefore equal the context baseline. Figure 1 shows the evaluated architecture and its deployment by scenario.

2.5 Separate spatial-adaptation extension

We also examine whether sparse target observations can improve regional reconstruction at unmonitored stations. This experiment uses a separate regional ExtraTrees product with 120 trees and 40 source stations selected for each target by environmental similarity. Source descriptors combine hydroclimatic summaries, ecology, coordinates, and network attributes. Selection and descriptor scaling use available covariates, not target DOC labels. Its predictors include the visible DOC-context features described above.

At each of 43 target stations, five observed months are reserved as support candidates: the first, middle, last, first-quartile, and third-quartile positions of the observed record. Nested prefixes give $K = 0, 1, 3, 5$ support budgets. All five candidates are excluded from evaluation at every budget, leaving the same 2,316 query cells. A shrunken mean support residual adjusts the fixed regional prediction:

$$\bar{r}_i^{(K)} = \frac{1}{K} \sum_{s \in \mathcal{S}_i^{(K)}} [z_{is} - \hat{z}_{is}^{\text{reg}}], \quad (7)$$

$$\hat{y}_{it}^{(K)} = \max\{0, \exp(\hat{z}_{it}^{\text{reg}} + \alpha_K \bar{r}_i^{(K)}) - 1\}. \quad (8)$$

The correction is zero for $K = 0$. The coefficients for $K = 1, 3, 5$ are 0.25, 0.50, and 0.75, inherited unchanged from an earlier internal station-validation analysis. The base predictions remain fixed.

Support can follow a query month, and station descriptors summarize the available historical record. This extension evaluates retrospective station reconstruction. Its source-model configuration and query set differ from the main hybrid experiment; it evaluates sparse calibration rather than the performance of a jointly trained hybrid-plus-adaptor model.

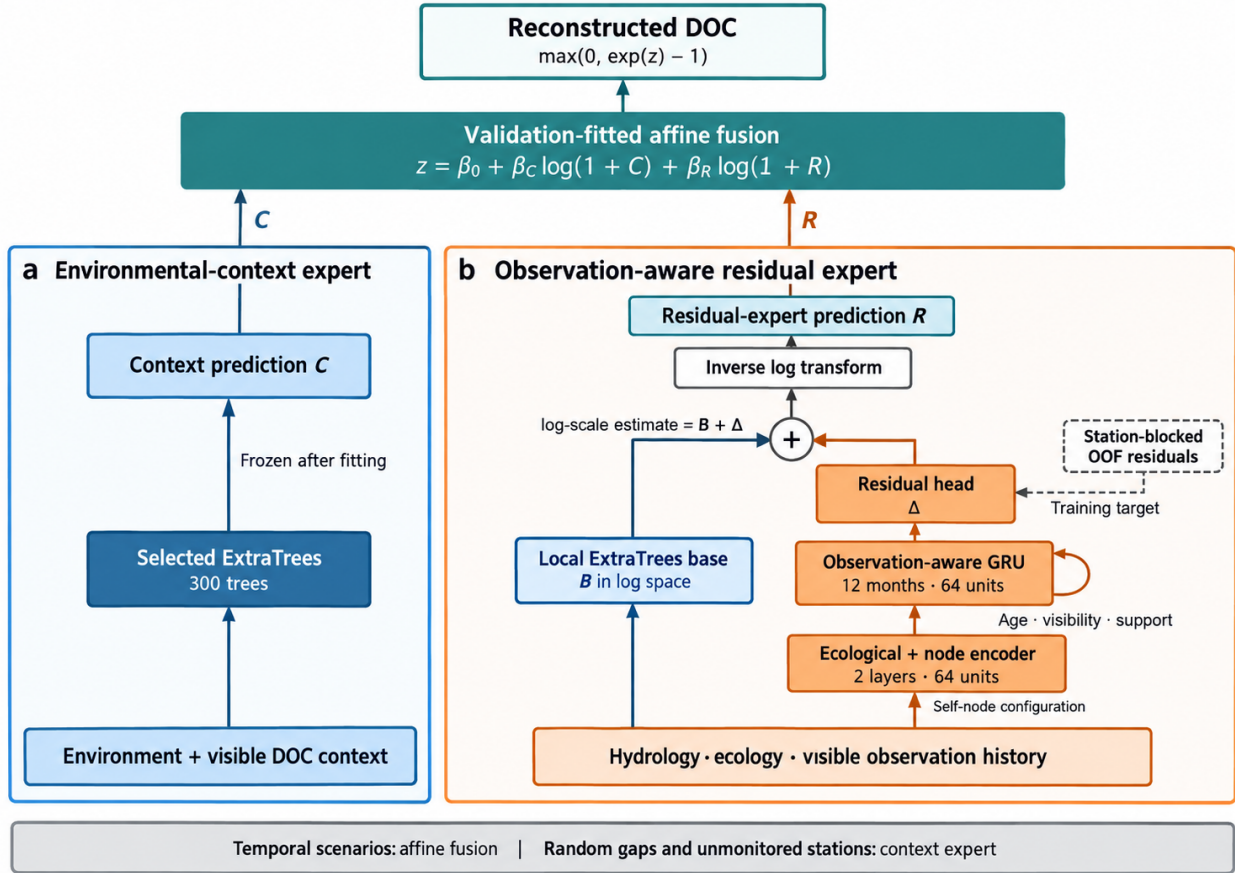


Figure 1: Environmental prediction and observation-aware residual learning. The context expert C and local reference B are separate tree models. The temporal branch learns out-of-fold residuals relative to B , producing the complete expert R after correction and inverse transformation. Validation-fitted affine fusion combines C and R for temporal scenarios. Random gaps and unmonitored stations use C . The selected encoder has a self-node configuration; learned river messages and attention are not part of this product. Dashed arrows indicate training supervision.

2.6 Evaluation and component comparisons

Primary performance is MAE on hidden DOC cells. RMSE, R^2 , log-space MAE, and training-derived Q90 tail MAE provide complementary summaries. Reported metrics average the five individual seed metrics; they are not the errors of a seed-ensemble prediction. The main table compares the fixed random-forest context model, validation-selected ExtraTrees context model, and final hybrid. In temporal scenarios, the complete residual expert is additionally compared on identical query cells.

For paired intervals, absolute errors are averaged across seeds at each cell, then whole stations are resampled 5,000 times. Each sampled station contributes all query cells, preserving the main cell-weighted estimand within each replicate. Intervals are reported for the hybrid-minus-context MAE difference and relative reduction. Seed variation is summarized separately. Q90 groups with fewer than 20 distinct cells are marked as small-sample diagnostics.

The spatial extension uses the same paired station-resampling principle. A descriptive station-shuffle diagnostic assigns fitted corrections to different target stations, using one fixed permutation per seed and support budget. Model development has repeatedly used these outer splits. Thus the reported evaluation quantifies performance on the existing development tasks; independent station partitions and external basins remain the next tests of generalization.

3 Results

3.1 The hybrid improves reconstruction across unobserved periods

The environmental-context predictor is a strong baseline across all four scenarios (Table 2). Combining it with the temporal residual expert reduces MAE from 1.047 to 0.887 mg/L in unobserved periods and from 1.016 to 0.784 mg/L in observation-assisted periods. These changes correspond to improvements of 15.3% and 22.9%, respectively. Both temporal scenarios improve across all five seeds.

The station-resampled 95% interval for the MAE difference is $[-0.262, -0.050]$ mg/L in unobserved periods and $[-0.307, -0.159]$ mg/L in observation-assisted periods. Corresponding relative-improvement intervals are 4.7–25.0% and 17.2–27.9% (Figure 2). The gains therefore persist when stations are resampled within each existing split.

Table 2: Five-seed reconstruction performance. MAE and RMSE are in mg/L. The hybrid equals the ExtraTrees context route in random gaps and unmonitored stations.

Scenario	RF MAE	ET MAE	Hybrid MAE	Hybrid RMSE	Hybrid R^2
Random gaps	1.452	1.436	1.436	6.626	0.384
Unobserved periods	1.070	1.047	0.887	1.404	0.470
Observation-assisted periods	1.051	1.016	0.784	1.228	0.587
Unmonitored stations	2.687	2.545	2.545	5.330	0.455

Random-gap MAE is 1.436 mg/L and unmonitored-station MAE is 2.545 mg/L, identical to the context expert by construction. These cases establish the retained baseline performance of the deployed configuration, rather than an increment from neural correction. Averaging losses across all scenario-cell occurrences gives hybrid MAE 1.475 mg/L, compared with 1.545 for ExtraTrees and 1.593 for the fixed random forest. The 11,046 scenario-cell occurrences include 8,150 distinct cells

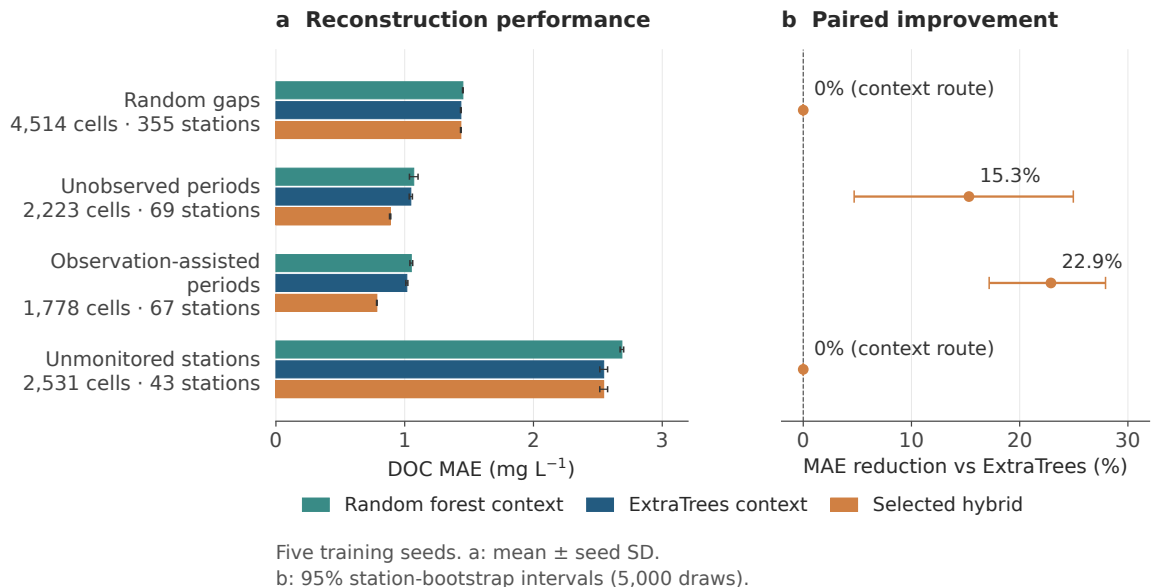


Figure 2: Performance by monitoring gap. **a**, Mean individual-seed MAE for matched context baselines and the hybrid; error bars show seed SD. **b**, Relative improvement over selected ExtraTrees with paired station-bootstrap 95% intervals. Zero gain in random gaps and unmonitored stations reflects the context-only route. Counts refer to unique query cells within each scenario.

because scenarios overlap; this pooled summary is descriptive, and the scenario-specific comparisons carry the main inference.

3.2 The two complete experts supply complementary predictions

The learned correction improves its matched local reference in both temporal scenarios (Table 3). In unobserved periods, MAE decreases from 1.148 mg/L for the local forest to 0.946 for the complete residual expert, a 17.6% reduction. In observation-assisted periods, it decreases from 0.973 to 0.795 mg/L, an 18.3% reduction. All five seeds improve in each comparison. This directly establishes the added value of the trained neural residual branch relative to its own base.

The complete residual expert also outperforms context ExtraTrees, whose MAEs are 1.047 and 1.016 mg/L. Affine fusion further reduces error to 0.887 and 0.784 mg/L, gains of 6.2% and 1.4% relative to the complete residual expert. All five seeds favor fusion in each comparison.

Table 3: Matched temporal component comparison. The local base is the forest corrected by the neural branch; the context expert is a separate forest. The complete residual expert includes its local base and correction. Values are mean individual-seed MAEs on identical query cells.

Scenario	Local base	Context	Residual expert	Hybrid
Unobserved periods	1.148	1.047	0.946	0.887
Observation-assisted periods	0.973	1.016	0.795	0.784

This sequence supports the complementary roles of the two complete predictors: the residual expert supplies complementary predictions, and the environmental-context expert still supplies in-

formation that improves their combined prediction. It does not isolate the individual contribution of every observation-age feature or identify a physical transport process. Earlier matched river-message and lag-attention controls gave little additional DOC improvement over self-node residual models, motivating the selected configuration. The main performance result is consequently attributed to the evaluated residual-expert combination.

3.3 Overall accuracy and high-concentration performance differ

Improvements in temporal MAE are accompanied by higher R^2 : from 0.384 to 0.470 in unobserved periods and from 0.428 to 0.587 in observation-assisted periods. RMSE decreases from 1.514 to 1.404 mg/L and from 1.446 to 1.228 mg/L, respectively. These changes extend the temporal gain beyond one loss metric.

High-concentration errors show a different pattern (Appendix A). The two temporal Q90 groups contain only 12 and seven distinct query cells. The hybrid changes tail MAE from 6.759 to 6.712 mg/L in unobserved periods and from 6.862 to 7.129 mg/L in observation-assisted periods. These small groups do not establish improved extreme-value reconstruction. For random gaps, selected ExtraTrees improves MAE slightly over the fixed random forest but has higher RMSE (6.626 versus 6.287 mg/L), further illustrating the importance of reporting more than average absolute error.

3.4 Sparse local calibration improves a separate spatial reconstruction

In the regional station-adaptation experiment, MAE decreases from 2.466 mg/L without target support to 2.320, 2.160, and 2.023 mg/L with one, three, and five observations per station (Table 4; Figure 3). Five observations yield a 17.9% reduction on the unchanged query. The paired MAE difference is -0.442 mg/L, with a station-resampled 95% interval of $[-1.026, -0.089]$ mg/L.

Table 4: Sparse station calibration in the separate regional experiment. Five seeds share 43 target stations and 2,316 fixed query cells. This baseline and query differ from Table 2.

K	MAE (mg/L)	Seed SD	RMSE (mg/L)	Reduction (%)
0	2.466	0.007	5.390	0.00
1	2.320	0.013	4.968	5.90
3	2.160	0.011	4.746	12.41
5	2.023	0.008	4.412	17.93

Calibration improves 36 of 43 stations. The median station-level relative reduction is 3.39%, smaller than the pooled 17.9%, indicating heterogeneous benefits and different weighting of station records (Figure 4). With five observations, assigning the station corrections to different targets increases mean MAE to 2.736 mg/L. This descriptive shuffle shows the importance of matching corrections to their own station; it does not identify the ecological cause of each station’s bias.

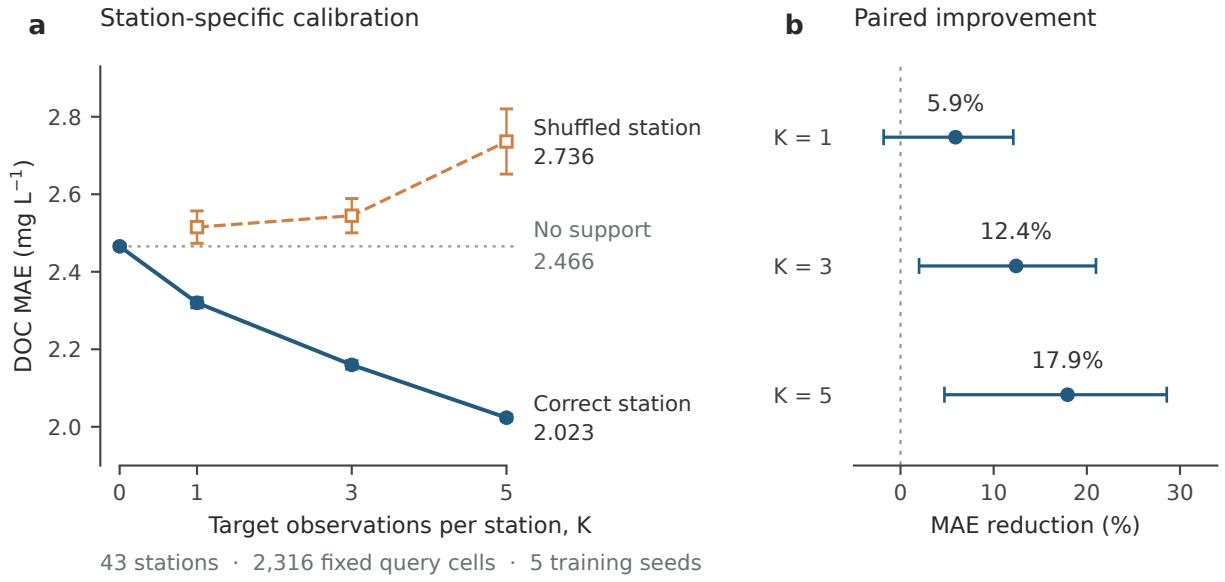
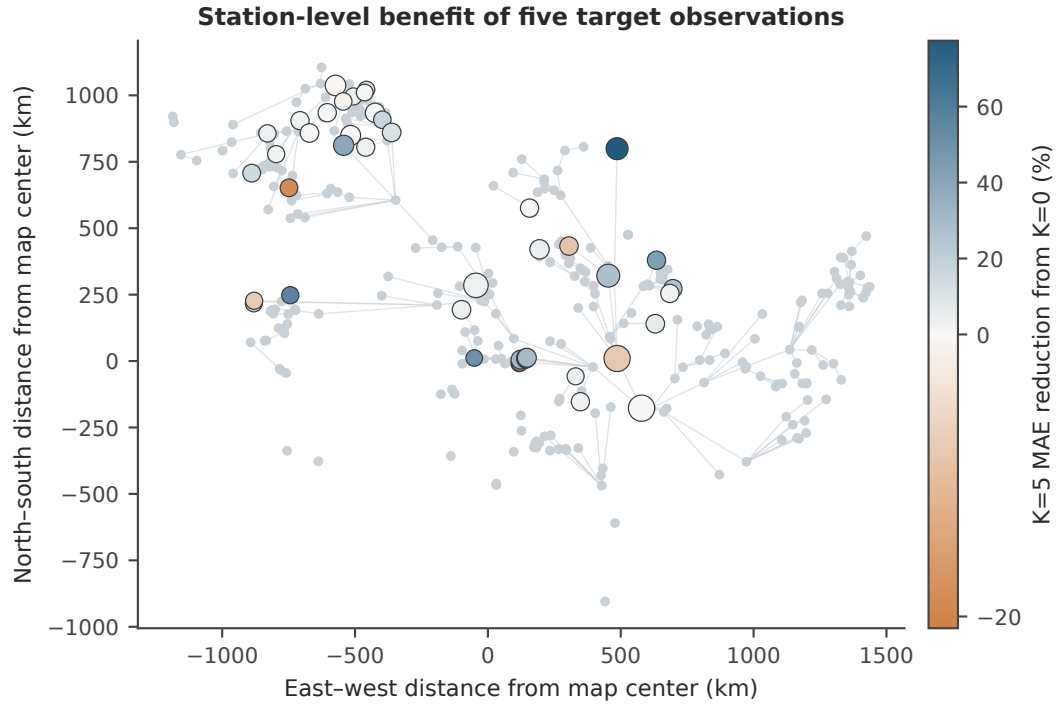


Figure 3: Sparse target observations improve regional DOC reconstruction. **a**, Mean DOC MAE across five training seeds as the number of observations per target station increases. The regional model and 2,316 query cells at 43 stations remain fixed. Correct-station calibration is compared with station-shuffled corrections and the uncalibrated reference; error bars show seed SD. Shuffling uses one fixed station permutation per seed and observation budget and serves as a descriptive diagnostic. **b**, Paired MAE reduction relative to no calibration, with 95% station-bootstrap intervals preserving cell weighting. Support may occur after query months, so this experiment evaluates historical reconstruction.



43 target stations; marker size represents query months.
 Grey links show the station graph, not river-channel geometry.
 Lambert azimuthal equal-area projection; center 96°W, 39°N.

Figure 4: Spatial heterogeneity in calibration benefit. Five-observation station responses from the same regional product used in Table 4. Marker size encodes the magnitude of relative MAE change. The projected station coordinates are real; connecting lines represent graph links rather than mapped river-channel geometry.

4 Discussion

4.1 Environmental prediction and temporal correction address different parts of the task

The results support a practical division of labor in sparse DOC reconstruction. An environmental-context forest supplies a strong reference using nonlinear environmental relationships and engineered observation summaries. The recurrent residual expert is trained to address errors left by a local environmental prediction. Fusing their complete outputs improves temporal reconstruction beyond either expert alone.

This design gives residual learning a specific role. Instead of requiring the recurrent network to estimate every part of DOC variation, the local forest anchors its prediction and the neural branch learns a correction. The out-of-fold construction makes this correction depend on errors at withheld training stations, reducing reliance on the forest’s fitted training residuals. Observation-aware memory provides a way to encode the changing availability of historical information. Our component comparison establishes complementarity at the expert level; it does not yet isolate whether elapsed observation time, ecological encoding, or recurrent state contributes most to that complementarity.

The methodological contribution is the organization and training of these complementary tasks for sparse DOC reconstruction. Tree ensembles, recurrent memory, observation decay, and residual modeling each have precedents. Their integration here is evaluated against tree models that already receive explicit lags and river-context features, making the observed temporal improvement a substantive prediction comparison rather than a consequence of withholding history from the baseline.

4.2 Different monitoring gaps call for different information

A station with a missing later record and a station with no target history present different learning problems. In the former, observations and hydroclimatic covariates can inform a time-dependent correction. In the latter, the model must transfer relationships across stations with different concentration levels and environmental conditions. The selected temporal hybrid improves the first task, whereas its deployed spatial route retains the context predictor.

The station-adaptation extension addresses a complementary spatial question. A small support set provides direct information about the target concentration level, and a station-specific correction improves the regional prediction without refitting its weights. The improvement across most held-out stations shows practical value, while its wide distribution shows that a small local sample does not help all stations equally. These findings motivate a future model that combines residual temporal learning and sparse spatial adaptation within one jointly evaluated workflow. The current experiments establish the two pieces separately.

4.3 Extending the framework with river transport

River-network geostatistics and graph imputation methods provide established ways to exploit relational structure [4, 7]. Explicit river-context summaries provide a strong source of spatial information in our context predictor. The earlier local-base message and lag-attention variants produced little additional improvement over their matched self-node residual controls. Their incremental value relative to the final selected context expert remains to be established.

A next step is to enrich the residual branch with flow-conditioned travel times, source–sink states, and observations along connected reaches. These additions would allow the model to repre-

sent how information moves through the river network as hydrological conditions change. Suitable flow, volume, and source information would also make it possible to test process constraints on transported loads. The residual formulation provides a direct way to evaluate these developments: each transport component can be tested for the additional information it supplies beyond the environmental reference and explicit river context.

4.4 Future research directions

The framework opens four connected directions for extending sparse DOC reconstruction. The first is transfer to new stations and river basins. Environmental similarity and a small number of target observations provide a starting point for adapting the model to different catchment settings. Future experiments should evaluate the fixed framework on new station groups and an independent basin, with inputs defined by the information available at deployment. This would identify which environmental relationships and temporal corrections transfer across regions and where local adaptation adds the most value.

The second direction is event-sensitive reconstruction. Higher-frequency DOC observations paired with rainfall and discharge would support learning of concentration changes during hydrological events. Event-weighted training and tail-focused objectives could then be compared for their ability to recover peak magnitude, timing, and duration. This development would connect the present monthly reconstruction to observed short-term carbon dynamics and provide a richer basis for evaluating high-concentration behavior.

The third direction is a unified model for spatial adaptation and temporal residual learning. Sparse local observations could update both the station-level concentration reference and the temporal correction, allowing adaptation to evolve as monitoring accumulates. A common evaluation should compare historical-record calibration with new-station adaptation using only earlier observations, while keeping the query cells fixed. This would bring the two capabilities demonstrated here into a single reconstruction workflow.

The fourth direction is uncertainty-aware river carbon assessment. Prediction intervals should be developed for the hybrid and evaluated jointly for coverage, width, and high-concentration performance. Combining reconstructed concentrations with discharge at a matching time scale would then support DOC-load estimation, with concentration and flow uncertainty propagated into the resulting loads. This would extend the framework from filling monitoring gaps toward quantifying carbon transport through river networks.

5 Conclusion

We developed and evaluated an environmental-context and observation-aware residual framework for sparse river DOC reconstruction. On the existing temporal holdouts, combining the two complete experts reduces MAE by 15.3% and 22.9% relative to selected ExtraTrees context predictions, with further improvement over the residual expert alone. These results show that a strong environmental reference and learned residual dynamics can supply complementary predictive information.

A separate spatial experiment demonstrates that five target observations can reduce regional reconstruction error by 17.9%, providing a practical extension toward poorly monitored stations. Together, these results provide a foundation for integrating temporal residual learning and local adaptation, extending reconstruction to new river basins, and developing event-sensitive river carbon assessments.

Data, code, and figure availability

The current manuscript is supported by local prediction products and reproducible analysis scripts. Main hybrid results are recomputed by `analyze_doc_hybrid_manuscript.py`; spatial-extension results are recomputed by `analyze_spatial_manuscript_evidence.py`. The scripts retain the source model, split, and prediction references used in each comparison. A public data and code release has not yet been assigned a persistent identifier. Figure 1 is an AI-assisted architecture schematic checked against the implemented model. All empirical plots are generated from saved numerical predictions.

A Tail diagnostics and implementation details

Table 5 records the high-concentration comparison without treating repeated predictions across seeds as additional observations. Thresholds are estimated from training labels separately for each scenario.

Table 5: Training-derived Q90 tail MAE (mg/L). Asterisks indicate fewer than 20 unique query cells.

Scenario	Threshold	Cells	ET MAE	Hybrid MAE
Random gaps	10.00	431	6.748	6.748
Unobserved periods	11.00	12*	6.759	6.712
Observation-assisted periods	11.00	7*	6.862	7.129
Unmonitored stations	9.40	510	6.906	6.906

The temporal expert uses nine ecological encoder inputs, 14 base monthly channels, and nine observation-aware channels before ecological embedding. The node representation has width 64 and the ecological embedding width 32. The self-node encoder is inherited from the graph implementation; excluding edge messages leaves the node transformation and explicit upstream-support feature intact. Residual targets, input visibility, and the terminal expert combination are defined by Equations 1–6. Reproduction should use the selected product settings described here, rather than earlier temporal models with different prediction clipping or training budgets.

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